BUILDING THE TWIN

Simulation II

Module 8 · A real-time twin with a camera in SUMO

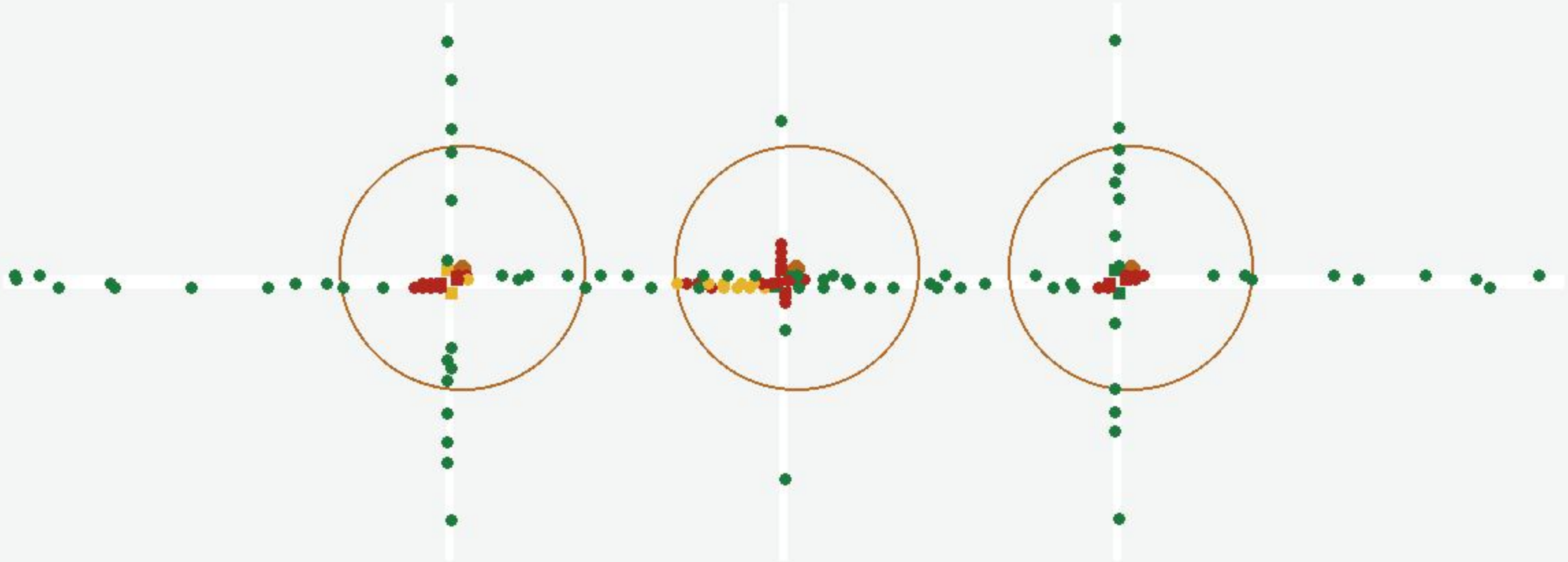
Transport Twin Study Path

The loop, every second



Watch the corridor

t = 898 s



A camera is not perfect

- Limited range: 110 m
- Misses: 98% close, 80% at the edge
- Occlusion: cars hide cars
- Noise and lane guessing

Held-out results

95 s vs 101 s

Time loss per trip at high demand: twin vs actuated (5 new seeds)

What we learned

Measure, don't assume

Sensing quality and decision logic both limit a twin